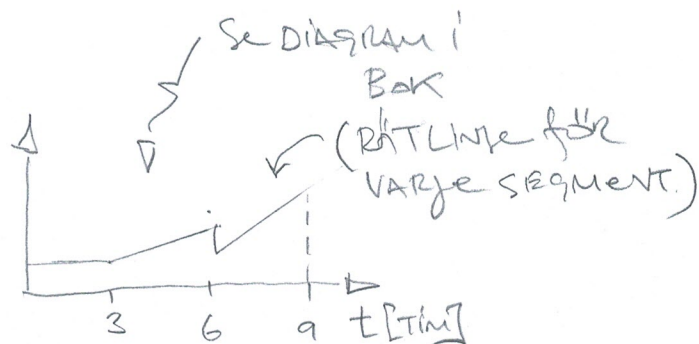
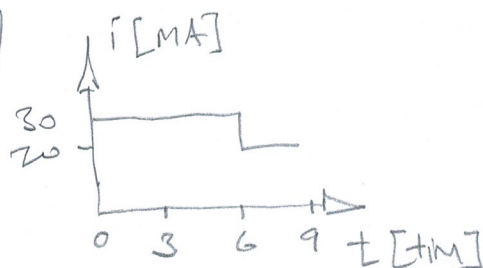


2.4)



a) TOTAL LADDNING?

VI VER ATT

$$i = \frac{dq}{dt} \text{ INTEGRERA } \Rightarrow \int i dt = \int dq$$

$$\Rightarrow Q = \int i dt = \int_0^{6 \cdot 3600} 30 \cdot 10^{-3} dt + \int_{6 \cdot 3600}^{9 \cdot 3600} 20 \cdot 10^{-3} dt =$$

$$= 684 + 216 = \underline{864 \text{ C}}$$

b) ENERGIN SOM BATTERIET ABSORBERAR?

$$VI \text{ VER ATT } P = \frac{dW}{dt} = iV \Rightarrow W = \int iV dt$$

FÖRSTÄMMENT.

$$W_1 = \int_0^{3600 \cdot 3} 30 \cdot 10^{-3} (0.5 + 9.259 \cdot 10^{-6} t) dt =$$

$$= 30 \cdot 10^{-3} \left[0.5t + \frac{1}{2} 9.259 \cdot 10^{-6} t^2 \right]_0^{3 \cdot 3600} =$$

$$= 178.1995 \approx \underline{178.2 \text{ J}}$$

$$W_2 = \int_{3 \cdot 3600}^{6 \cdot 3600} 30 \cdot 10^{-3} (5.55 \cdot 10^{-5} t) dt$$

$$= 30 \cdot 10^{-3} \left[\frac{1}{2} 5.55 \cdot 10^{-5} t^2 \right]_{3 \cdot 3600}^{6 \cdot 3600} =$$

$$= 291.3084 \approx \underline{291.31 \text{ J}}$$

$$W_3 = \int_{6.3600}^{9.3600} 20 \cdot 10^{-3} (-1.9 + 1.11 \cdot 10^{-4} t) dt =$$

$$= 20 \cdot 10^{-3} \left[-1.9t + \frac{1}{2} 1.11 \cdot 10^{-4} t^2 \right]_{6.3600}^{9.3600} =$$

$$= 236.952 \approx \underline{236.95} \text{ J}$$

SUMMIERAS

$$W = W_1 + W_2 + W_3 = \underline{706.46} \text{ J}$$

SVAR

a) $Q = 864 \text{ C}$

b) $W = 706.5 \text{ J}$